

Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1. Generating a the sequence from 1-55, increasing by fives.
```

```
my_seq <- seq(1, 55, 5)
my_seq
```

```
## [1] 1 6 11 16 21 26 31 36 41 46 51
```

```
#2. Mean and median of this sequence
```

```
mean_seq <- mean(my_seq)
median_seq <- median(my_seq)
mean_seq
```

```
## [1] 26
```

```
median_seq
```

```
## [1] 26
```

```
#3. Determine whether the mean is greater than the median
mean_seq > median_seq
```

```
## [1] FALSE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
#5. Create three vectors with four components each
student_names <- c("James", "Martin", "Jackson", "Mark") # Character vector for student names
test_scores <- c(80, 91, 95, 88) # Numeric vector for test scores
scholarship <- c(FALSE, TRUE, TRUE, FALSE) # Logical Vector for scholarship status

#7. Combine the vectors into a data frame.
student_info <- data.frame(student_names, test_scores, scholarship)

#8. Label the columns.
colnames(student_info) <- c("Student_Name", "Test_Score", "Scholarship_status")
student_info # print student_info
```

```
##   Student_Name Test_Score Scholarship_status
## 1      James         80             FALSE
## 2     Martin         91              TRUE
## 3    Jackson         95              TRUE
## 4       Mark         88             FALSE
```

9. QUESTION: How is this data frame different from a matrix?

Answer: A matrix can only hold one type of data, usually numeric, but this data frame is more like a spread sheet that could contain different data types.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

#10. Create a function using if...else

```
pass_fail_if <- function(x) {  
  if (x > 50) {  
    return("Pass")  
  } else {  
    return("Fail")  
  }  
}
```

#11. Create a function using ifelse()

```
pass_fail_ifelse <- function(x) {  
  ifelse(x > 50, "Pass", "Fail")  
}
```

#12a. Run the first function with the value 52.5

```
pass_fail_if(52.5)
```

```
## [1] "Pass"
```

#12b. Run the second function with the value 52.5

```
pass_fail_ifelse(52.5)
```

```
## [1] "Pass"
```

#13a. Run the first function with the vector of test scores

```
#pass_fail_if(test_scores)
```

Doesn't work properly "Error in if (x > 50) { : the condition has length > 1"

#13b. Run the second function with the vector of test scores

```
pass_fail_ifelse(test_scores) # Works with the vector test_scores
```

```
## [1] "Pass" "Pass" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The “ifelse” function worked. According to Yale Center for Research Computing: “Most of R’s functions are vectorized, meaning that the function will operate on all elements of a vector without needing to loop through and act on each element one at a time.” Function “ifelse” is vectorized so it worked for the complete vector of `test_scores`

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)